

Activity 1: DFA Design

Activity 1: DFA Design – Even Number of 1s

Problem Statement:

Design a Deterministic Finite Automaton (DFA) that accepts all binary strings over $\Sigma = \{0,1\}$ which contain an even number of 1s.

Student Tasks:

1. Define the DFA as
 $M = (Q, \Sigma, \delta, q_0, F)$
2. Draw the state transition diagram.
3. Identify accepting and rejecting states.
4. Test the DFA for inputs:
 - o 1010
 - o 111
 - o 1100

Answers:

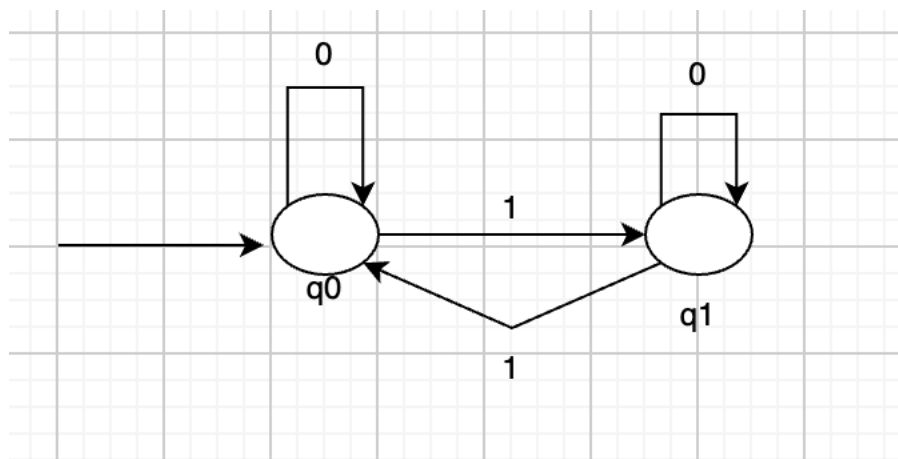
1. Define the DFA as

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where:

- $Q = \{q_0, q_1\}$
- $\Sigma = \{0, 1\}$
- q_0 = Start state
- $F = \{q_0\}$ (Accepting state)
- δ = Transition function

2. Draw the state transition diagram.



3. Identify accepting and rejecting states.

- **Accepting State:** q_0 (Even number of 1s)
- **Rejecting State:** q_1 (Odd number of 1s)

4. Test the DFA for inputs:

Input: **1010**

Input	State
Start	q_0
1	q_1
0	q_1
1	q_0
0	q_0

Final State: q_0

Result:  Accepted (2 ones – even)

Input: **111**

Input	State
Start	q_0
1	q_1
1	q_0
1	q_1

Final State: q_1

Result: Rejected (3 ones – odd)

Input: **1100**

Input	State
Start	q_0
1	q_1
1	q_0
0	q_0
0	q_0

Final State: q_0

Result:  Accepted (2 ones – even)

Result

The DFA successfully accepts all binary strings containing an **even number of 1s** and rejects strings containing an **odd number of 1s**

Activity 2: DFA for Strings Ending with “01”

Problem Statement:

Construct a DFA over $\Sigma = \{0,1\}$ that accepts all strings that end with “01”.

Student Tasks:

1. Design the DFA with minimum number of states.
2. Draw the transition diagram.
3. Write the transition table.
4. Trace the DFA for inputs:

- o 01
- o 101
- o 1101
- o 100

1. DFA Definition

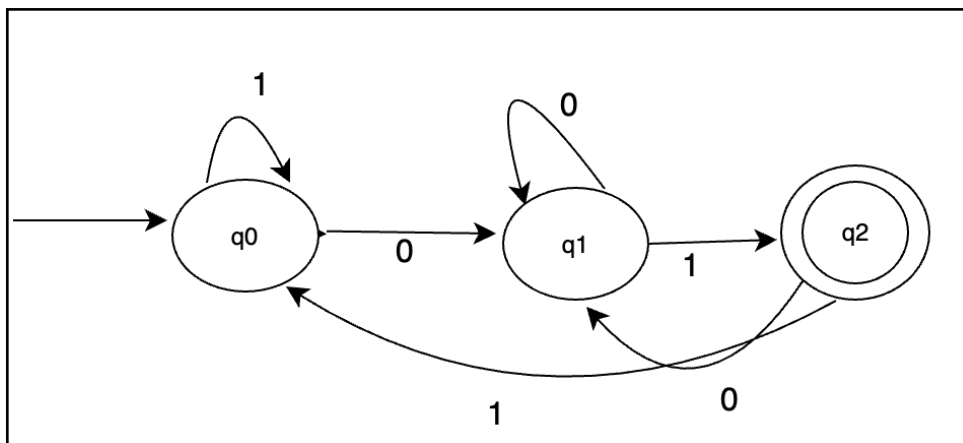
Let

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where:

- $Q = \{q_0, q_1, q_2\}$
- $\Sigma = \{0,1\}$
- q_0 = Start state
- $F = \{q_2\}$
- δ = Transition function

2. Transition Diagram



3. Transition Table

Present State	Input 0	Input 1
$\rightarrow q_0$	q_1	q_0

q ₁	q ₁	q ₂
*q ₂	q ₁	q ₀

4. DFA Trace

Input: **01**

Current State	Input	Next state
q ₀	0	q ₁
q ₁	1	q ₂

Final State: q₂  **Accepted**

Input: 101

Current State	Input	Next state
q ₀	1	q ₀
q ₀	0	q ₁
q ₁	1	q ₂

Final State: q₂  **Accepted**

Input: **1101**

Current State	Input	Next state
q ₀	1	q ₀
q ₀	1	q ₀
q ₀	0	q ₁
q ₁	1	q ₂

Final State: q₂  **Accepted**

Input: **100**

Current State	Input	Next state
q0	1	q0
q0	0	q1
q1	0	q1

Final State: q₁ **✗ Rejected**

Number of States: 3

Start State: q₀

Final State: q₂

Accepted Strings: Strings ending with **01**

Rejected Example: 100 (does not end with 01)

Activity 3: Identify Accept/Reject from Given DFA

Given a DFA that accepts string over $\Sigma = \{a, b\}$ which contain at least one 'a' over the string

Tasks:

1. Identify the final (Ga) state.
2. For the following inputs, say Accept or Reject and show the transition path:
 - bbbb
 - abbb
 - bab
 - ϵ (empty string)
3. Write the language accepted by the DFA in words.

1. Identify the Final (Accepting) State

Final State: q₁

2. Accept/Reject and Transition Path

(a) Input: bbbb

Transition Path:

q₀ → q₀ → q₀ → q₀ → q₀

Final State: q₀

Result: Reject

(b) Input: abbb

Transition Path:

$q_0 \rightarrow q_1 \rightarrow q_1 \rightarrow q_1 \rightarrow q_1$

Final State: q_1

Result: Accept

(c) Input: bab

Transition Path:

$q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow q_1$

Final State: q_1

Result: Accept

(d) Input: ϵ (Empty String)

Transition Path:

q_0

Final State: q_0

Result: Reject

3. Language Accepted by the DFA

Language:

All strings over the alphabet $\{a, b\}$ that contain at least one a.

Accepted Language:

$L = \{ w \in \{a, b\}^* \mid w \text{ contains at least one } a \}.$

Activity 4: Real-Time Application DFA – Password Rule

Design a DFA that accepts all strings over $\Sigma = \{a, b\}$ that Start with 'a' and End with 'b'

Tasks:

1. Draw the DFA.
2. Clearly mark start and final states.
3. Check whether the following are accepted or rejected:
 - ab
 - aab
 - abb
 - ba

- a

1. DFA Definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where:

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{a, b\}$$

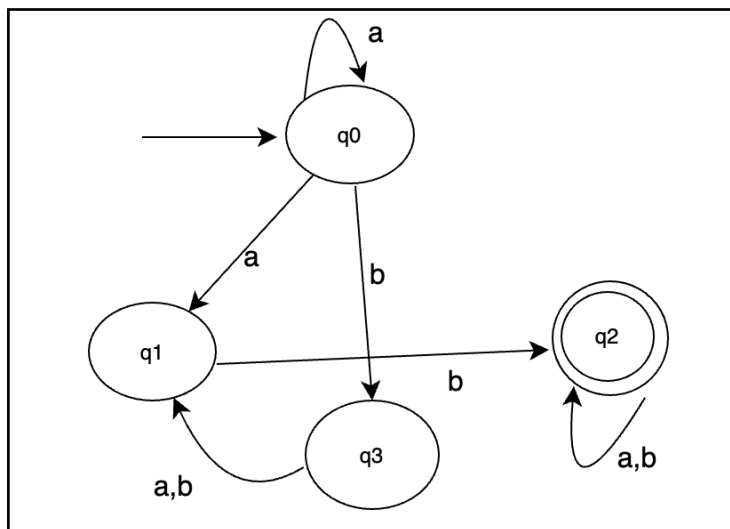
q_0 = Start state

$$F = \{q_2\}$$

2. Transition Table

Present State	Present State	Present State
q_0	q_0	q_0
q_1	q_1	q_2
$*q_2$	q_1	q_2
q_3	q_3	q_3

3. Transition Diagram



4. Check the Given Inputs

Input: ab

Path:

$q_0 \rightarrow q_1 \rightarrow q_2$

Result: Accepted

Input: aab

Path:

$q_0 \rightarrow q_1 \rightarrow q_1 \rightarrow q_2$

Result: Accepted

Input: abb

Path:

$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_2$

Result: Accepted

Input: ba

Path:

$q_0 \rightarrow q_3 \rightarrow q_3$

Result: Rejected

Number of States: 4

Start State: q_0

Final State: q_2

Accepted Strings: All strings over $\{a, b\}$ that start with a and end with b